

Discrete defect filters and source-resolved positivity for Erdős–Straus

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Abstract

The original ES return operator has reciprocal odd-integer spectrum, not an arbitrary interval spectrum. Explicit interpolation filters give one-sided counting bounds with factorial error and coefficient norm growing only linearly. They yield uniform, unscaled count certificates of moment degree $O(\log p / \log \log p)$, with all source and split-fibre marks retained. An exact hard-prime shell shows that occupied shells need not be detected by the previous aggregate first-degree lower bound. A source-resolved estimate and a fixed quartic filter both recover its positive integral witness; a second hard-prime example also defeats separate-channel degree one. An explicit all-integer arithmetic progression follows from the recovered selector. No universal ES theorem or historical-priority conclusion is asserted.

1 Original arithmetic and what the uniform estimate leaves to prove

The source is the preceding infinite-array tranche and its pinned finite SplitZero application [1, 2]. We retain all original labels, not only the eigenvalues of their observation. Let $p = 4m + 1$ be prime. The canonical row consists of

$$m + 1 \leq a \leq 2m, \quad R = 4a - p, \quad a = \prod_i q_i^{e_i}, \quad -e_i \leq \beta_i \leq e_i, \quad c \in \{E, M\}. \quad (1)$$

Set

$$u = \prod_i q_i^{e_i + \beta_i}, \quad g_E = 4u + 1, \quad g_M = u + a, \quad D_\alpha = \frac{R}{\gcd(R, g_c)}, \quad x_\alpha = D_\alpha^{-1}. \quad (2)$$

Every original prime-power defect and channel remains part of α . The exact possible observation values are

$$\Sigma = \{0\} \cup \{1/(2k + 1) : k \geq 0\}, \quad x_\alpha \in \Sigma \setminus \{0\}. \quad (3)$$

Zero is a possible spectral limit, not an absent labelled state. The inherited bound treated this set as the larger interval union $[0, 1/3] \cup \{1\}$. The present improvement uses the gaps inside that interval.

For completeness, the positive rational candidate is

$$E : \left(a, \frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right), \quad M : \left(a, \frac{p(a + a^2/u)}{R}, \frac{p(a + u)}{R} \right). \quad (4)$$

The two nonfirst reduced denominators are both exactly D_α . Indeed $(RY - pa)(RZ - pa) = p^2a^2$, and $4a \equiv p \pmod{R}$ with $\gcd(pa, R) = 1$ reduces the numerator gcds to those in (2). Consequently a positive integral witness is exactly a state with $D_\alpha = 1$. These are classical Type I/II coordinates, with the marking retained [3, §2, Propositions 2.2 and 2.6].

Writing $d_0 = \gcd(a, u)$ gives the complete normalization

$$(h, r, s) = (d_0^2/u, u/d_0, a/d_0), \quad a = hrs, \quad u = hr^2, \quad \gcd(r, s) = 1.$$

At a hit the quotient and raw ordered denominators are

$$E : \kappa = (pr + s)/R, \quad (a, hsk, phrk), \quad M : \lambda = (r + s)/R, \quad (a, phs\lambda, phr\lambda). \quad (5)$$

The ordered inverse recovers $u = a^2/(RY - pa)$ for E and $u = pa^2/(RY - pa)$ for M, then recovers every $\beta_i = v_{q_i}(u) - e_i$. There is no unmarked interchange of the last two denominators.

Why the row is complete. Here is the previously established first-half argument, included to identify its exact hypothesis. Order a positive integral solution as $x \leq y \leq z$. Then $p/4 < x \leq 3p/4 < p$. If $x > p/2$, the reciprocal equation implies $y < p$. Thus $z = pk$ by clearing denominators modulo p . Put $t = (4k - 1)/p \in \mathbb{Z}_{>0}$; it has $t \equiv 3 \pmod{4}$, $\gcd(t, k) = 1$, and $txy = k(x + y)$. Write $x = du, y = dv$ with $\gcd(u, v) = 1, u \leq v$. Then $uv \mid k$, so $k = uvl$ and $td = l(u + v)$. Coprimality gives $u + v = th, d = lh$. The inequality $2x > p$ now gives $2lu(u - v) > -1$, forcing $u = v = 1$. But then $t \mid 2$, contrary to $t \equiv 3 \pmod{4}$. Equality $x = p/2$ is impossible for odd p . For $a = x$, both other denominators exceed $p/2$, and the positive integer factor pair $(RY - pa, RZ - pa)$ has product $p^2 a^2$. Its two p -adic orders are $(0, 2)$, $(1, 1)$, or $(2, 0)$, yielding exactly (4), with a recorded exterior reversal in the last case. Thus (1) decides existence without restricting a ray or a Koide chamber.

Let W_p and H_p be the canonical row's total and successful counts. The original counting Hilbert space and projector are

$$\mathcal{H} = \ell^2\left(\bigsqcup_p \mathcal{A}_p\right), \quad Xe_\alpha = x_\alpha e_\alpha, \quad \Pi e_\alpha = \mathbf{1}_{D_\alpha=1} e_\alpha. \quad (6)$$

The preceding theorem gives $\|X^n - \Pi\| \leq 3^{-n}$ and compensates row mass by an explicit prime-dependent power. It does not itself prove $H_p > 0$. The following construction yields one-sided inequalities with faster convergence and a controlled moment-coefficient norm.

2 Discrete interpolation filters and their exact constants

For $j, s \geq 0$ define

$$P_j(x) = \prod_{\nu=1}^j \frac{(2\nu+1)x-1}{2\nu}, \quad F_{j,s}(x) = x^s P_j(x), \quad \Phi_j = F_{j,1}. \quad (7)$$

These are explicit rational polynomials. In particular Φ_j is the unique polynomial of degree at most $j+1$ with values one at 1 and zero at $0, 1/3, \dots, 1/(2j+1)$. The full evaluation map on these $j+2$ distinct nodes is an isomorphism; its inverse is

$$(v_i)_i \mapsto \sum_i v_i \prod_{h \neq i} \frac{x - z_h}{z_i - z_h}. \quad (8)$$

This is the classical Lagrange interpolation map [4, §3.3(i)]. Only its stated inverse column is used in (7); no novelty of interpolation itself is claimed.

Theorem 1 (Exact one-sided filters). *For $D = 2k + 1 > 1$,*

$$F_{j,s}(1/D) = \begin{cases} 0, & 1 \leq k \leq j, \\ (-1)^j \frac{\binom{k-1}{j}}{(2k+1)^{j+s}}, & k > j. \end{cases} \quad (9)$$

Also $F_{j,s}(1) = 1$, and its coefficient ℓ^1 norm is exactly $j+1$. Thus even j gives an upper and odd j a lower bound for the indicator of $D = 1$. On the entire set (3),

$$\boxed{\Phi_{2r+1} \leq \mathbf{1}_{\{1\}} \leq \Phi_{2r}, \quad \|\Phi_j - \mathbf{1}_{\{1\}}\|_\Sigma \leq \frac{1}{2^j j! (2j+3)}}. \quad (10)$$

The even sequence decreases and the odd sequence increases to that indicator.

Proof. Each factor at $D = 2k + 1$ is $(\nu - k)/(\nu D)$, proving (9) and the sign. For $k > j$, $\binom{k-1}{j}/D^j \leq 1/(2^j j!)$ and $D \geq 2j + 3$. This proves (10), including the value zero at the accumulation point. All nonzero coefficients of P_j alternate in sign, so their absolute sum is $|P_j(-1)| = \prod_{\nu=1}^j (\nu+1)/\nu = j+1$; multiplication by x^s only shifts them. For the same-parity comparison, the additional two factors have product between zero and one at every remaining failed node, and are one at the successful node. Nodes newly encountered become zero. These statements prove monotonicity and convergence. $\square$

The sign assertion is only on the actual discrete spectrum. For example $P_2(1/4) < 0$; using it as an upper bound on all of $[0, 1/3]$ would be false. Even return denominators are excluded by the arithmetic proof, not by fiat.

Theorem 2 (A finite formula for the sharp hybrid error). *For $j \geq 1, s \geq 1$, the exact maximum error of $F_{j,s}$ is obtained among*

$$k \in \mathbb{Z}, \quad \max\{j+1, \lfloor 1 + j(j+2)/(2s) \rfloor\} \leq k \leq \max\{j+1, \lceil j + j(j+2)/(2s) \rceil\}. \quad (11)$$

Evaluate the positive rational expression in (9) at these finitely many integers and take its maximum. For $j = 0, s > 0$, the maximum is 3^{-s} . For $s = 0$ the supremum is $1/(2^j j!)$, approached as $D \rightarrow \infty$.

Proof. For real $k > j$, logarithmic differentiation of $\binom{k-1}{j}/(2k+1)^{j+s}$ gives the sign of

$$\sum_{\nu=1}^j \frac{2\nu+1}{k-\nu} - 2s.$$

The sum is strictly decreasing from infinity to zero. Its unique crossing is between $1 + j(j+2)/(2s)$ and $j + j(j+2)/(2s)$, since $\sum (2\nu+1) = j(j+2)$. Before that crossing the function increases and after it decreases. The maximizing integer, also respecting $k \geq j+1$, lies in (11). The boundary cases follow directly from (9). $\square$

In particular the first errors for Φ_j , $j = 0, 1, 2, 3$, are $1/3, 2/49, 10/2197, 4/9261$, attained at $D = 3, 7, 13, 21$. The theorem is a maximization on the full allowed odd spectrum; it does not assert that each maximizing D occurs in every arithmetic row.

3 Uniform row positivity, original mass and certified precision

For any finite original selection A , put

$$M_k(A) = \sum_{\alpha \in A} D_\alpha^{-k}, \quad Q_j(A) = \sum_{\alpha \in A} \Phi_j(D_\alpha^{-1}).$$

The count H_A retains repeated states with different original labels. Theorem 1 proves

$$\boxed{Q_{2r+1}(A) \leq H_A \leq Q_{2r}(A), \quad |Q_j(A) - H_A| \leq \frac{|A|}{2^j j!(2j+3)}}. \quad (12)$$

The same formulas allow positive prescribed masses, but integer recovery below uses the original unit masses, not a row normalized to total mass one.

The original row bound is

$$W_p \leq B_p = \min\{6m^2, 4m(1 + \lfloor \log_2(2m) \rfloor)^2\}. \quad (13)$$

Indeed $\tau(a^2) \leq 2a - 1$ gives the first expression. Prime by prime, $2e+1 \leq (e+1)(e+2)/2$, so $\tau(a^2) \leq d_3(a)$. Then $\sum_{a \leq N} d_3(a) \leq N(\sum_{u \leq N} 1/u)^2$ and dyadic grouping bounds the harmonic sum by $1 + \lfloor \log_2 N \rfloor$. This proves (13).

Theorem 3 (Uniform unscaled lower certificates). *For $n \geq 0$, let $j(p, n)$ be the least even nonnegative integer satisfying*

$$2^j j!(2j+3) \geq 8 \cdot 3^n B_p. \quad (14)$$

Set $\ell_{p,n} = Q_{j+1}(\mathcal{A}_p)$ and $u_{p,n} = Q_j(\mathcal{A}_p)$. Simultaneously for every row,

$$\boxed{H_p - \frac{1}{16 \cdot 3^n} \leq \ell_{p,n} \leq H_p \leq u_{p,n} \leq H_p + \frac{1}{8 \cdot 3^n}}. \quad (15)$$

In particular $\lceil \ell_{p,n} \rceil = \lfloor u_{p,n} \rfloor = H_p$, and

$$\boxed{H_p > 0 \iff \ell_{p,n} > 0}. \quad (16)$$

The largest moment index is $j(p, n) + 2$. For fixed n , it is $(1 + o(1)) \log p / \log \log p$ as $p \rightarrow \infty$ through these rows.

Proof. Apply (12)–(14). The ratio of consecutive denominators is $2(j+1)(2j+5)/(2j+3) > 2$, proving the lower error in (15). The bracket has width at most $3/(16 \cdot 3^n) < 1$ and contains the integer H_p . For $H_p \geq 1$ its lower endpoint is positive; for $H_p = 0$ it is nonpositive. All degree choices depend only on the proved mass bound, not on success. Finally the integral bounds for $\sum_{h=1}^j \log h$ give $\log(2^j j! (2j+3)) = j \log j + O(j)$. Minimality among even integers changes this by only $O(\log j)$. Since $\log B_p = \log p + O(\log \log p)$, the asserted degree estimate follows. $\square$

This is a certificate-degree improvement, not a bound on the cost of computing the original moments. For a mass bound 10^{12} the stated schedule needs moments through degree 14; for 10^{24} , through degree 22. At the first hard prime of Section 5, its whole-row mass bound is 22 829 467 930 080, giving maximum moment degree 16. Its whole row was not enumerated in this contribution.

Absolute input error. If every supplied moment through degree $j+2$ has certified absolute error at most $\delta = (16(j+2)3^n)^{-1}$, polynomial evaluation error is at most $(j+1)\delta$ for the upper filter and $(j+2)\delta$ for the lower. These follow from the exact coefficient norm in Theorem 1. Enlarge both endpoints in the required directions. The resulting interval still contains H_p and has width less than $7/(16 \cdot 3^n) < 1$. Original masses are unchanged; if moments are large, obtaining this absolute precision still requires the corresponding relative precision.

Infinite operators. On the original countable space (6), the pointwise inequalities give $\Phi_{2r+1}(X) \preceq \Pi \preceq \Phi_{2r}(X)$ with the same operator-norm errors. Let $Ne_p = H_p e_p$ on the maximal domain $\{v : \sum H_p^2 |v_p|^2 < \infty\}$. The row operators with diagonals $\ell_{p,n}, u_{p,n}$ have that same domain; their differences from N are bounded with the constants in (15). This follows directly by coordinate norms, so no unproved trace of an infinite identity is used. For real $\sigma > 1$, the signed relative row traces converge absolutely, bounded by the corresponding constant in (15) times $\sum_p p^{-\sigma}$.

For the whole programme, (16) gives

$$\forall p \ H_p > 0 \iff \inf_p \ell_{p,0} > 0 \iff \inf_p \ell_{p,0} \geq 15/16. \quad (17)$$

This is a precise forcing target, not a proof of the last bound. An empty row would have every odd-filter sum nonpositive. The present argument does not rule out such a row merely by assembling all rows.

4 Source-resolved lower bounds and exact transport

For a genuine partition $A = \bigsqcup_{\lambda} A_{\lambda}$, retain its original unit masses. Let $L_d(\mu_{\lambda})$ be the previous signed variational bound

$$L_d(\mu_{\lambda}) = \max_{\deg P \leq d, P(1)=1} \frac{3}{2} \int (x - 1/3) P(x)^2 d\mu_{\lambda}. \quad (18)$$

Its continuous-support sign is valid on $[0, 1/3] \cup \{1\}$; the discrete polynomials (7) use the stronger exact spectrum and belong to a different family. We do not claim one family dominates the other at every degree.

Proposition 4 (A lower bound retaining the partition). *The integer*

$$C_d(\{A_{\lambda}\}) = \sum_{\lambda} \max\{0, \lceil L_d(\mu_{\lambda}) \rceil\} \quad (19)$$

is a lower bound for H_A . Refining the original partition cannot decrease it. In particular a positive score on one original channel certifies an ES witness without requiring a positive sum after merging the channels.

Proof. Each $L_d(\mu_{\lambda}) \leq H_{A_{\lambda}}$ and the latter is a nonnegative integer. For a refinement of one part, every common polynomial has score the sum of its refined scores, so $L_{\text{coarse}} \leq \sum L_{\text{fine}}$. Use $\lceil \sum x_i \rceil \leq \sum \lceil x_i \rceil$ and discard negative summands to obtain (19) and its monotonicity. $\square$

At degree one, the two-channel coefficient space has block form $\text{diag}(J_E, J_M)$. Restriction through $\Delta P = (P, P)$ has form $J_E + J_M$. This is restriction to a particular subspace, not an equivalence of the forms. The inverse of Δ on its image reads either identical entry; the difference component must be retained for the full direct sum.

Split arithmetic. For a budget split $e = f + b$, the addition map has fibre

$$\max(-f_i, \beta_i - b_i) \leq \xi_i \leq \min(f_i, \beta_i + b_i), \quad \eta = \beta - \xi,$$

with cardinality $n(\beta)$ equal to the product of those interval lengths. Assign each occurrence weight $1/n(\beta)$. For every new filter and every moment, the occurrence sum equals the original sum. This is proved by grouping the actual finite fibres, not by assuming an unweighted convolution. In the corresponding completed spaces, duplication U and fibre averaging $E = U^*$ satisfy $EU = 1$. Since $\Phi_j(0) = 0$,

$$\boxed{\Phi_j(UXU^*) = U\Phi_j(X)U^*}. \quad (20)$$

Its inverse on $\text{im } U$ uses U^* , and the complementary kernel is retained. The raw repeated diagonal operator still counts split multiplicities; (20) is its compressed transport, not a licence to take the raw trace. For P_j without the extra x , the missing term would be $(-1)^j (2^j j!)^{-1} (I - UU^*)$.

Every polynomial observation retains its source state α and D_α . Forgetting them can merge zeroed defects $3, 5, \dots, 2j + 1$. A scalar moment vector does not reconstruct which divisor supplied a hit. Positive lower scores imply that some original summand is positive; enumerating its finite original fibre then returns a $D = 1$ label and (5). The prime-power socle and augmentation maps commute with these blockwise scalar polynomials, without turning supported zeros into external absence.

5 An actual hard-prime obstruction and two positive repairs

Take

$$\boxed{\begin{aligned} p_* &= 19\,748\,674\,681 \equiv 1 \pmod{840}, & R &= 135, \\ a_* &= 4\,937\,168\,704 = 2^6 \cdot 13^4 \cdot 37 \cdot 73. \end{aligned}} \quad (21)$$

This is a prime, certified by the full order criterion with base 11 and

$$p_* - 1 = 2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 23510327.$$

All factors are proved prime by complete trial division. The residues of $11^{(p_*-1)/q}$, in this factor order, are

$$19748674680, 10039250736, 3428156451, 13907516733, 12674160518.$$

Each displayed residue minus one is coprime to p_* , and $11^{p_*-1} = 1 \pmod{p_*}$. For any prime divisor l of p_* , these identities force the order modulo l to be divisible by $p_* - 1$, hence $l \geq p_*$. This proves primality. The program separately tests every possible odd divisor through 140529.

The following table exhausts the 1053 divisors and both original channels:

D	1	3	5	9	15	27	45	135
E	1	39	53	80	69	144	244	423
M	0	40	54	80	68	144	244	423
$E + M$	1	79	107	160	137	288	488	846

(22)

It can be independently reproduced in the finite group algebra by multiplying

$$\left(\sum_{i=0}^{12} [2]^i\right) \left(\sum_{j=0}^8 [13]^j\right) \left(\sum_{k=0}^2 [37]^k\right) \left(\sum_{l=0}^2 [73]^l\right) \quad \text{in } \mathbb{Z}[(\mathbb{Z}/135\mathbb{Z})^\times] \quad (23)$$

and applying the two gcd gates to each residue. The independent direct exponent enumeration is also supplied, with all 2106 labels. No group-generated support is substituted for these bounded exponents.

For $S_i = \sum_\alpha \gcd(R, g_\alpha)^i$, the first values are

$$(S_0, S_1, S_2, S_3, S_4) = (2106, 13962, 315738, 12455226, 722183418).$$

Let $A_i = 3S_{i+1} - RS_i$. Then

$$(A_0, A_1, A_2) = (-242424, -937656, -5258952), \quad A_0 A_2 - A_1^2 = 395697405312 > 0. \quad (24)$$

The degree-one signed localizing matrix is congruent, by a positive diagonal coordinate change, to $\begin{pmatrix} A_0 & A_1 \\ A_1 & A_2 \end{pmatrix}$. It is negative definite. Thus the shell has a hit but *every* aggregate degree-one normalized polynomial fails to give a positive score. Its exact maximum is $-50887012/144801015$. This is an obstruction to extending the earlier finite observation, not a counterexample to ES or a correction of the proved sufficient criterion. No leastness is claimed.

First repair: retain the channel. On E alone, the fixed polynomial $P(x) = (36x - 1)/35$ has score

$$\boxed{\frac{3}{2} \sum_{\alpha \in E} (x_\alpha - 1/3) P(x_\alpha)^2 = \frac{64376}{196875} > 0.} \quad (25)$$

The exact E maximum is $1692013/5172810 > 0$; the M maximum is negative. Thus (19) forces at least one original E witness.

Second repair: exploit the discrete spectrum. The fixed quartic filter

$$\Phi_3(x) = \frac{105x^4 - 71x^3 + 15x^2 - x}{48}$$

gives, on the original *combined* state set,

$$\boxed{Q_3 = \frac{1297691}{2460375} > 0, \quad Q_4 = \frac{115374406}{110716875}.} \quad (26)$$

Their integer bracket is exactly $H = 1$. In particular, the arithmetic inequality

$$\boxed{105S_4 - 71RS_3 + 15R^2S_2 - R^3S_1 > 0} \quad (27)$$

is a valid sufficient condition for a hit in *every* original shell. It is satisfied here although (24) prevents first-degree certification. This uses one higher moment for positivity, with no singular-Gram inverse.

The unique original hit in this shell is

$$(c, u, h, r, s, \kappa) = (E, 141879296, 26, 2336, 81289, 341725215823),$$

$$(x, y, z) = (4937168704, 722241027794932022, 409884191461517484986392768). \quad (28)$$

The centered vector is $(5, -3, -1, 1)$. Its slope $2336/81289 < 1/8$ also belongs to the prescribed lower positive chamber. That chamber assertion is only for this marking, not every later member of the progression below.

Channel separation alone is not asserted universal. There is a second fully certified hard prime

$$p_{**} = 2671180768668258904496300170662649 \equiv 289 \pmod{840},$$

with $a = (p_{**} + 135)/4 = 2^5 \cdot 431^3 \cdot 457^3 \cdot 7229^4$. Its 9702 canonical candidates contain three E hits, but

$$L_1(E + M) = -\frac{7138170316}{1971871105}, \quad L_1(E) = -\frac{146243172}{493377905}, \quad L_1(M) < 0.$$

The fixed discrete quartic nevertheless gives

$$Q_3 = \frac{54448414}{66430125} > 0.$$

The exact full histogram, three witnesses and recursive Lucas certificate are in the JSON. The eight nontrivial certificate nodes reduce to trial-prime leaves below 10000; no probable-prime assumption is used. This additional case excludes a premature claim that channel separation always suffices.

6 A constructive infinite arithmetic specialization

The first recovered witness extends pointwise, without requiring its parameter to be prime. For every integer $j \geq 0$, put

$$t = 73 + 945j, \quad C = 67632448, \quad n = 4Ct - 135 = 19748674681 + 255650653440j,$$

$$h_0 = 26, \quad r_0 = 32t, \quad s_0 = 81289, \quad a = Ct,$$

$$\kappa_0 = 341725215823 + 8847406287328j + 57265746370560j^2.$$

Polynomial expansion proves $135\kappa_0 = nr_0 + s_0$ and $a = h_0r_0s_0$. Therefore

$$\boxed{\frac{4}{n} = \frac{1}{a} + \frac{1}{h_0s_0\kappa_0} + \frac{1}{nh_0r_0\kappa_0}}. \quad (29)$$

All denominators are positive integers for every $j \geq 0$. If $d = \gcd(r_0, s_0)$, then $\gcd(d, 135) = 1$ and $d \mid \kappa_0$. The canonical return is

$$(h, r, s, \kappa) = (h_0d^2, r_0/d, s_0/d, \kappa_0/d), \quad u = h_0r_0^2,$$

which leaves the ordered denominators unchanged. The actual inverse parameter is $j = (n - 19748674681)/255650653440$ on this image. Every n in (29) is $1 \pmod{840}$, and the initial term is coprime to the step. Dirichlet's theorem therefore gives infinitely many prime members [6, Theorem 18.1]. Formula (29) already proves ES for every integer member. This is a specialization of the inherited selector, not a claim that arithmetic-progression constructions are new or that this progression exhausts the hard locus.

7 Verification, source scope and next pointwise obligation

The standalone deterministic standard-library verifier proves the two primes by exact certificates, enumerates both specified shells in full, compares direct and residue-algebra censuses, verifies the moment inequalities and their positive witness inverses, and tests the entire finite node window used for sharp errors. It also checks interpolation inverse columns, interval precision, completed split trace identities and a full small-prime row comparison. Its finite output is not a proof of the infinite estimates; those were proved above.

The original infinite-array and SplitZero source measures, kernels and labels are retained. No theta-source norm, Frobenius action, Leech geometry or cubic surface existence theorem is imported to assert positivity of these arithmetic sums. The generic interpolation and moment principles have their primary citations. The application and examples were calculated here; a scoped search did not establish historical priority.

The uniform estimates now give explicit original-moment separating inequalities, with factorial rather than merely geometric error and controlled coefficient size. Two actual hard-prime shells show why a fixed first-degree shortcut cannot be inferred from the earlier census, and how stronger signed estimates recover positive witnesses. What is not proved is that every canonical row defined here, for primes $p \equiv 1 \pmod{4}$, must satisfy one of these positive lower inequalities. The remaining universal assertion is the arithmetic lower bound in (17), not an unproved limit interchange.

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